Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 & Machine Learning Lab		
Academic year	2026-2027 (Even)	Batch:2023-2027	**Due date:** 28-12-2025

Experiment 1: Working with Python packages-Numpy, Scipy, Scikit-Learn, Matplotlib

Aim: To study and explore the fundamental Python libraries used in data science and machine learning, and to understand how different datasets can be analyzed and mapped to appropriate machine learning models using exploratory data analysis techniques.

Libraries used:

- **NumPy**: Used for numerical computations and efficient handling of multi-dimensional arrays.
- **Pandas**: Used for data manipulation, cleaning, and analysis using DataFrames.
- **Matplotlib**: Used for creating visualizations such as line graphs, bar charts, and histograms.
- **Seaborn**: Used for advanced statistical data visualization with attractive and informative plots.

Objectives performed:

1 Loan Amount Prediction

[32]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("datasets/loan_train.csv")
print("Head (First 5 rows)\n", df.head())
print("Shape\n", df.shape)
```

```
Head (First 5 rows)
   Customer ID               Name Gender  Age  Income (USD) Income Stability  \
0     C-36995   Frederica Shealy      F   56       1933.05              Low
1     C-33999  America Calderone      M   32       4952.91              Low
2      C-3770      Rosetta Verne      F   65        988.19             High
3     C-26480         Zoe Chitty      F   65           NaN             High
4     C-23459       Afton Venema      F   31       2614.77              Low

  Profession     Type of Employment    Location  Loan Amount Request (USD)  \
0    Working            Sales staff  Semi-Urban                   72809.58
1    Working                    NaN  Semi-Urban                   46837.47
2  Pensioner                    NaN  Semi-Urban                   45593.04
```

```
3  Pensioner                     NaN       Rural                    80057.92
4    Working  High skill tech staff  Semi-Urban                   113858.89

   ...  Credit Score No. of Defaults Has Active Credit Card  Property ID  \
0  ...        809.44               0                     NaN          746
1  ...        780.40               0            Unpossessed          608
2  ...        833.15               0            Unpossessed          546
3  ...        832.70               1            Unpossessed          890
4  ...        745.55               1                 Active          715

   Property Age  Property Type Property Location  Co-Applicant  \
0       1933.05              4             Rural             1
1       4952.91              2             Rural             1
2        988.19              2             Urban             0
3           NaN              2        Semi-Urban             1
4       2614.77              4        Semi-Urban             1

   Property Price  Loan Sanction Amount (USD)
0       119933.46                    54607.18
1        54791.00                    37469.98
2        72440.58                    36474.43
3       121441.51                    56040.54
4       208567.91                    74008.28

[5 rows x 24 columns]
Shape
 (30000, 24)
```

[33]:
```python
print("Info \n", df.info())
print("Describe \n", df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 24 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   Customer ID                  30000 non-null  object
 1   Name                         30000 non-null  object
 2   Gender                       29947 non-null  object
 3   Age                          30000 non-null  int64
 4   Income (USD)                 25424 non-null  float64
 5   Income Stability             28317 non-null  object
 6   Profession                   30000 non-null  object
 7   Type of Employment           22730 non-null  object
 8   Location                     30000 non-null  object
 9   Loan Amount Request (USD)    30000 non-null  float64
 10  Current Loan Expenses (USD)  29828 non-null  float64
 11  Expense Type 1               30000 non-null  object
```

```
 12  Expense Type 2               30000 non-null  object
 13  Dependents                   27507 non-null  float64
 14  Credit Score                 28297 non-null  float64
 15  No. of Defaults              30000 non-null  int64
 16  Has Active Credit Card       28434 non-null  object
 17  Property ID                  30000 non-null  int64
 18  Property Age                 25150 non-null  float64
 19  Property Type                30000 non-null  int64
 20  Property Location            29644 non-null  object
 21  Co-Applicant                 30000 non-null  int64
 22  Property Price               30000 non-null  float64
 23  Loan Sanction Amount (USD)   29660 non-null  float64
dtypes: float64(8), int64(5), object(11)
memory usage: 5.5+ MB
Info
 None
Describe
                 Age  Income (USD)  Loan Amount Request (USD)  \
count  30000.000000  2.542400e+04               30000.000000
mean      40.092300  2.630574e+03               88826.333855
std       16.045129  1.126272e+04               59536.949605
min       18.000000  3.777000e+02                6048.240000
25%       25.000000  1.650457e+03               41177.755000
50%       40.000000  2.222435e+03               75128.075000
75%       55.000000  3.090593e+03              119964.605000
max       65.000000  1.777460e+06              621497.820000

       Current Loan Expenses (USD)    Dependents  Credit Score  \
count                 29828.000000  27507.000000  28297.000000
mean                    400.936876      2.253027    739.885381
std                     242.545375      0.951162     72.163846
min                    -999.000000      1.000000    580.000000
25%                     247.667500      2.000000    681.880000
50%                     375.205000      2.000000    739.820000
75%                     521.292500      3.000000    799.120000
max                    3840.880000     14.000000    896.260000

       No. of Defaults   Property ID  Property Age  Property Type  \
count     30000.000000  30000.000000  2.515000e+04   30000.000000
mean          0.193933    501.934700  2.631119e+03       2.460067
std           0.395384    288.158086  1.132268e+04       1.118562
min           0.000000      1.000000  3.777000e+02       1.000000
25%           0.000000    251.000000  1.650450e+03       1.000000
50%           0.000000    504.000000  2.223250e+03       2.000000
75%           0.000000    751.000000  3.091408e+03       3.000000
max           1.000000    999.000000  1.777460e+06       4.000000

       Co-Applicant  Property Price  Loan Sanction Amount (USD)
```

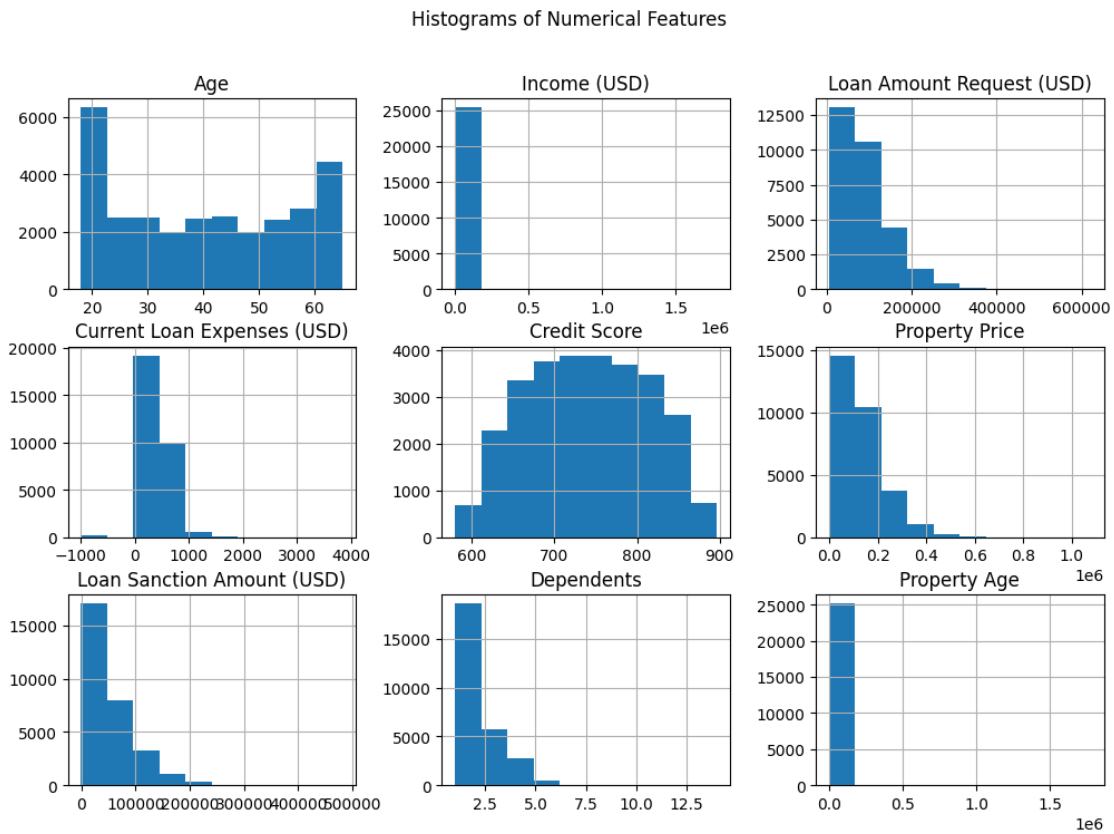
```
count  30000.000000    3.000000e+04                29660.000000
mean      -4.743867    1.317597e+05                47649.342208
std       74.614593    9.354955e+04                48221.146686
min     -999.000000   -9.990000e+02                 -999.000000
25%        1.000000    6.057216e+04                    0.000000
50%        1.000000    1.099936e+05                35209.395000
75%        1.000000    1.788807e+05                74261.250000
max        1.000000    1.077967e+06               481907.320000
```

[34]:
```
print(df.isnull().sum())
```

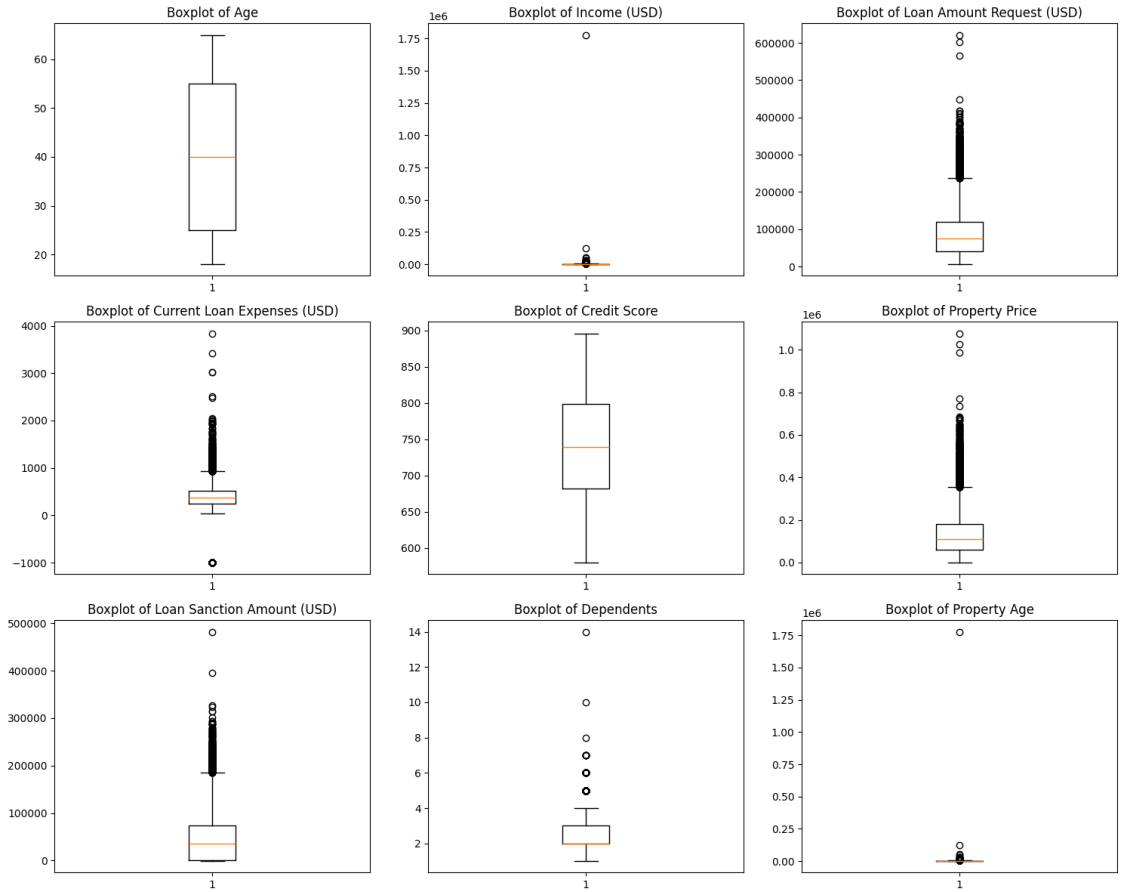
```
Customer ID                      0
Name                             0
Gender                          53
Age                              0
Income (USD)                  4576
Income Stability              1683
Profession                       0
Type of Employment            7270
Location                         0
Loan Amount Request (USD)        0
Current Loan Expenses (USD)    172
Expense Type 1                   0
Expense Type 2                   0
Dependents                    2493
Credit Score                  1703
No. of Defaults                  0
Has Active Credit Card        1566
Property ID                      0
Property Age                  4850
Property Type                    0
Property Location              356
Co-Applicant                     0
Property Price                   0
Loan Sanction Amount (USD)     340
dtype: int64
```

Exploratory Data Analysis

[35]:
```
numcols = ['Age', 'Income (USD)', 'Loan Amount Request (USD)',
'Current Loan Expenses (USD)', 'Credit Score',
'Property Price', 'Loan Sanction Amount (USD)','Dependents','Property Age']

df[numcols].hist(figsize=(12,8))
plt.suptitle("Histograms of Numerical Features")
plt.show()
```



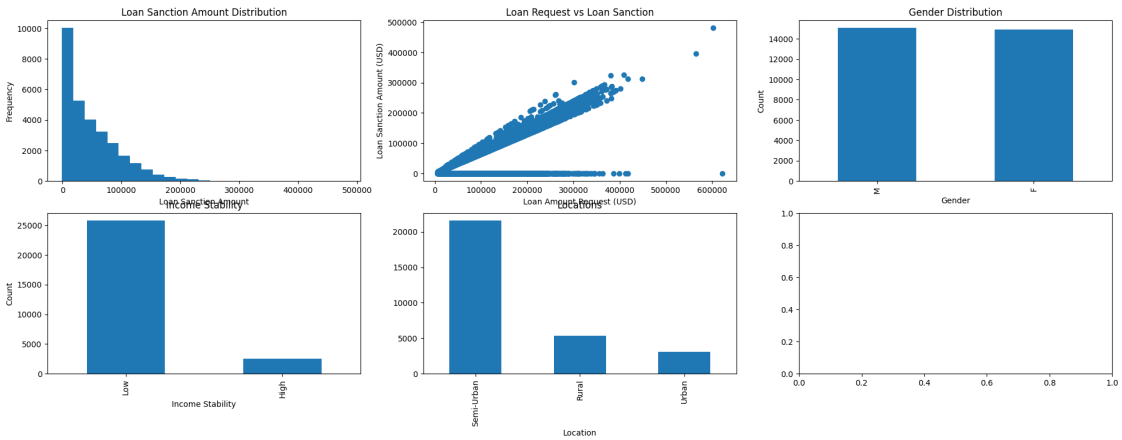
[36]:
```
import matplotlib.pyplot as plt

num_cols = [
    'Age', 'Income (USD)', 'Loan Amount Request (USD)',
    'Current Loan Expenses (USD)', 'Credit Score',
    'Property Price', 'Loan Sanction Amount (USD)',
    'Dependents','Property Age'
]

fig, axes = plt.subplots(3, 3, figsize=(15, 12))  # 3x3 grid
axes = axes.flatten()  # flatten to easily loop

for ax, col in zip(axes, num_cols):
    ax.boxplot(df[col].dropna())
    ax.set_title(f"Boxplot of {col}")

plt.tight_layout()
plt.show()
```



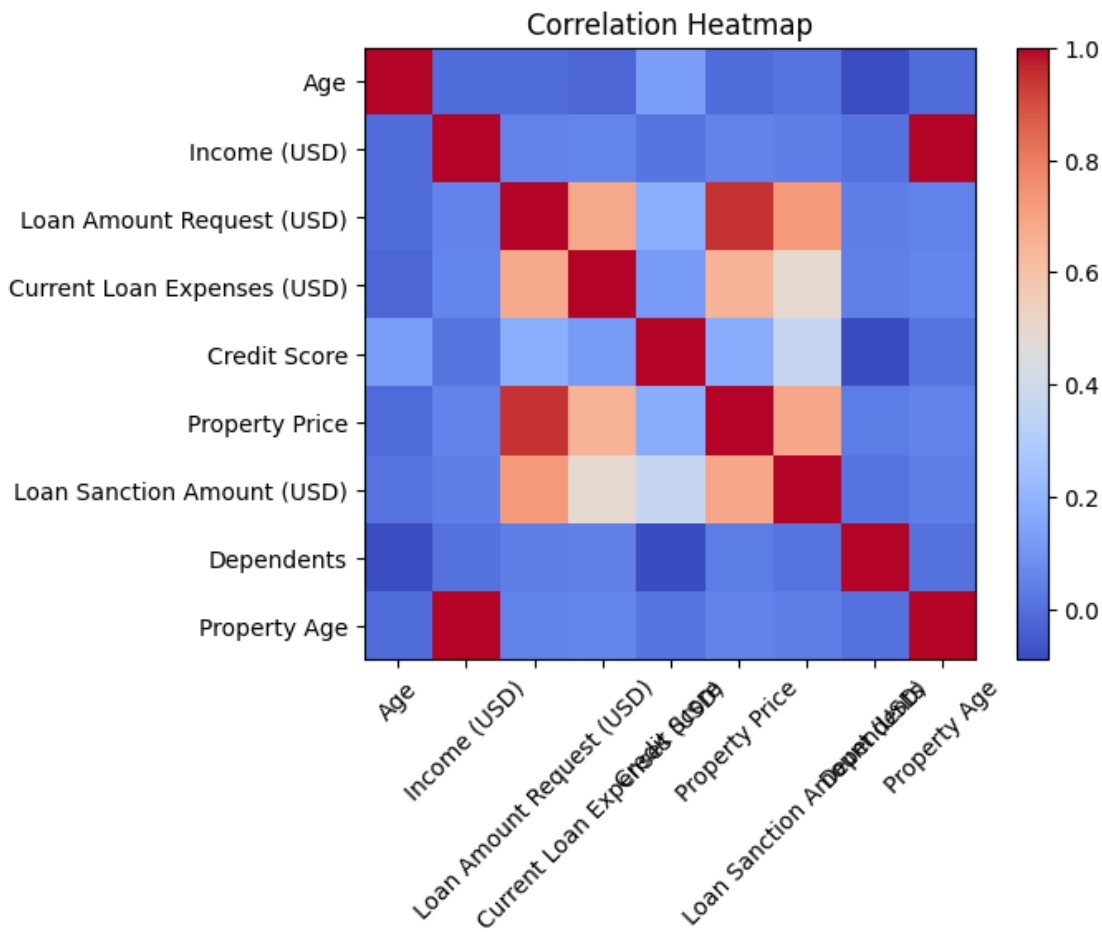
[37]:
```
fig, axes = plt.subplots(2, 3, figsize=(24, 8))
axes = axes.flatten()
# Histogram
axes[0].hist(df["Loan Sanction Amount (USD)"], bins=25)
axes[0].set_title("Loan Sanction Amount Distribution")
axes[0].set_xlabel("Loan Sanction Amount")
axes[0].set_ylabel("Frequency")

axes[1].scatter(df["Loan Amount Request (USD)"], df["Loan Sanction Amount␣
↪(USD)"])
axes[1].set_xlabel("Loan Amount Request (USD)")
axes[1].set_ylabel("Loan Sanction Amount (USD)")
axes[1].set_title("Loan Request vs Loan Sanction")

df["Gender"].value_counts().plot(kind = 'bar', ax = axes[2])
axes[2].set_title("Gender Distribution")
axes[2].set_xlabel("Gender")
axes[2].set_ylabel("Count")
```

```
df["Income Stability"].value_counts().plot(kind = 'bar', ax = axes[3])
axes[3].set_title("Income Stability")
axes[3].set_xlabel("Income Stability")
axes[3].set_ylabel("Count")

df["Location"].value_counts().plot(kind = 'bar', ax = axes[4])
axes[4].set_title("Locations")
axes[3].set_ylabel("Count")
```

[37]: Text(0, 0.5, 'Count')



[38]:
```
corr = df[numcols].corr()
plt.imshow(corr, cmap='coolwarm')
plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=45)
plt.yticks(range(len(corr.columns)), corr.columns)
plt.title("Correlation Heatmap")
plt.show()
```



2 Iris

[39]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

df=pd.read_csv('datasets/Iris.csv')
df.head()
```

[39]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

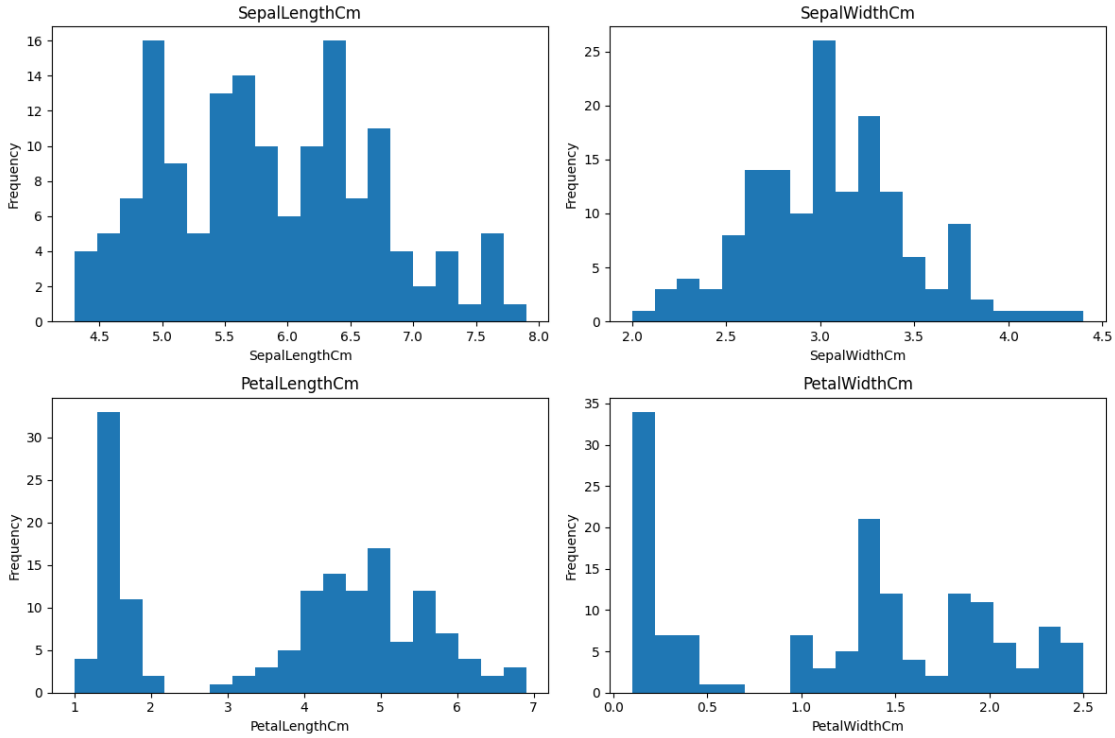
[40]:
```
df.describe()
```

[40]:

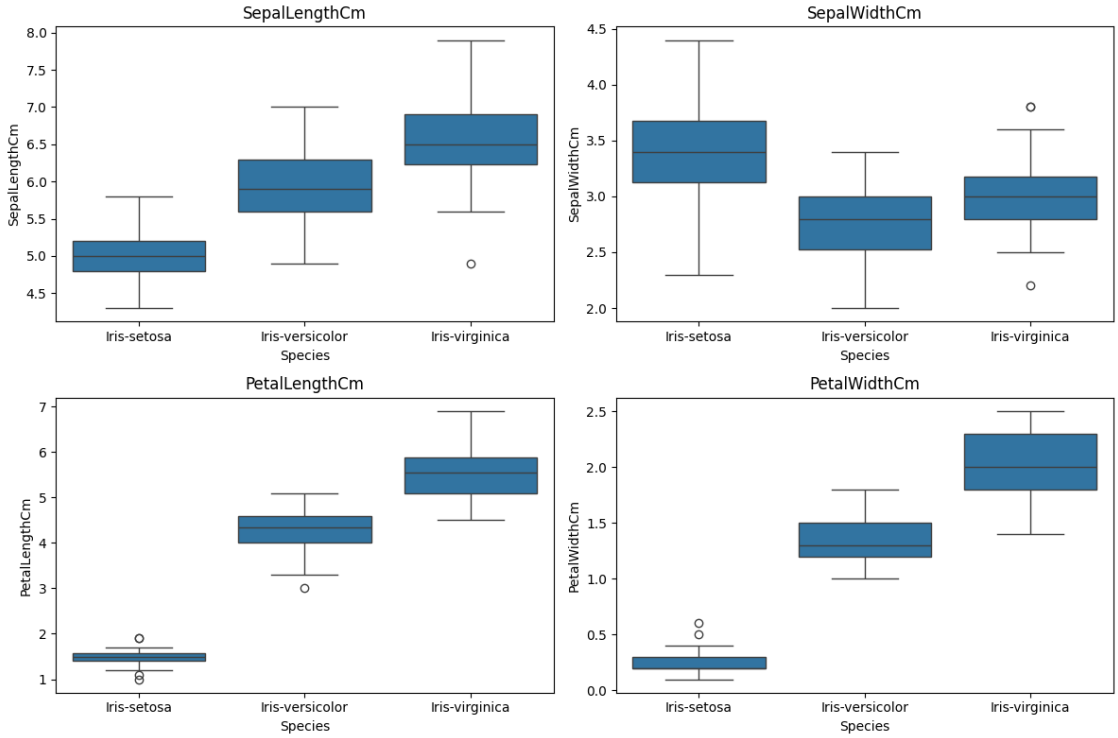
	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

[41]:

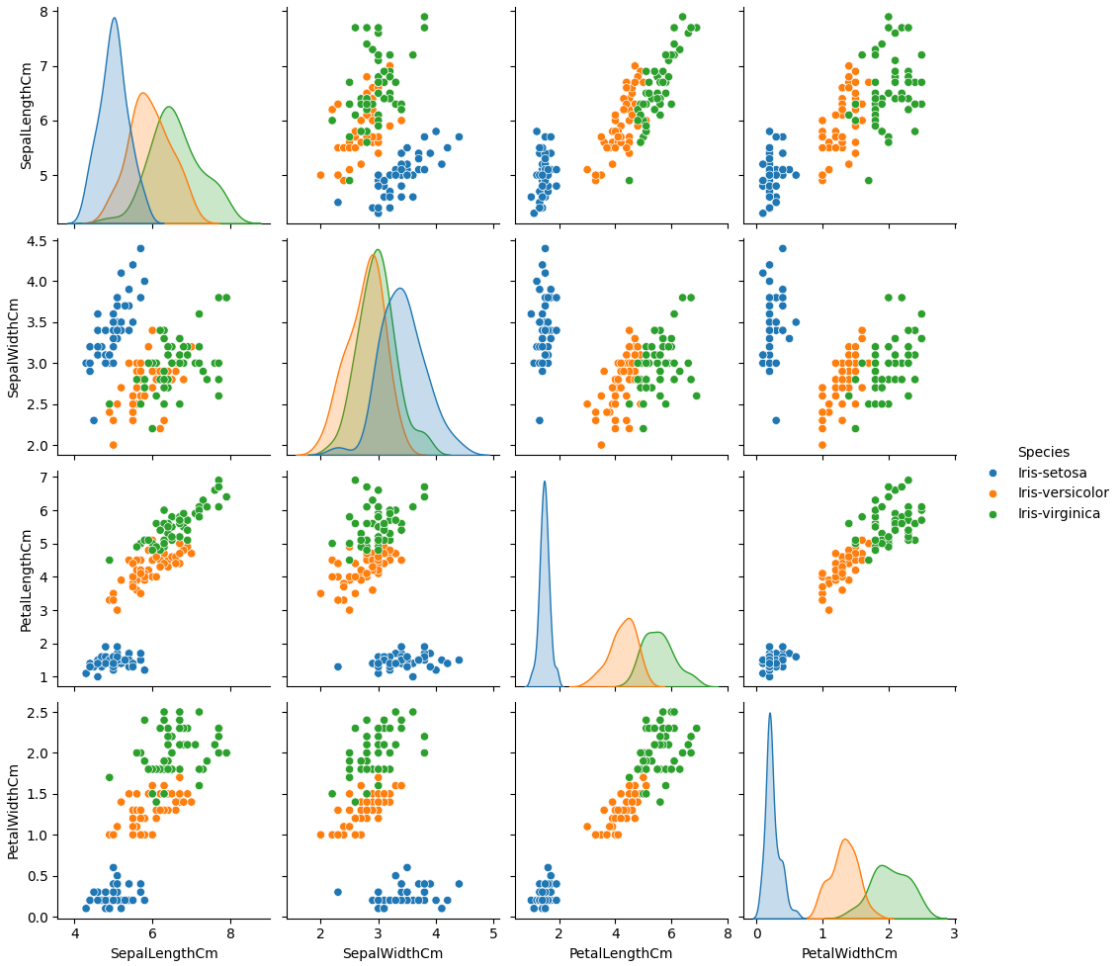
```
features = [
    'SepalLengthCm',
    'SepalWidthCm',
    'PetalLengthCm',
    'PetalWidthCm'
]

plt.figure(figsize=(12, 8))

for i, col in enumerate(features, 1):
    plt.subplot(2, 2, i)
    plt.hist(df[col], bins=20)
    plt.title(col)
    plt.xlabel(col)
    plt.ylabel('Frequency')

plt.tight_layout()
plt.show()
```



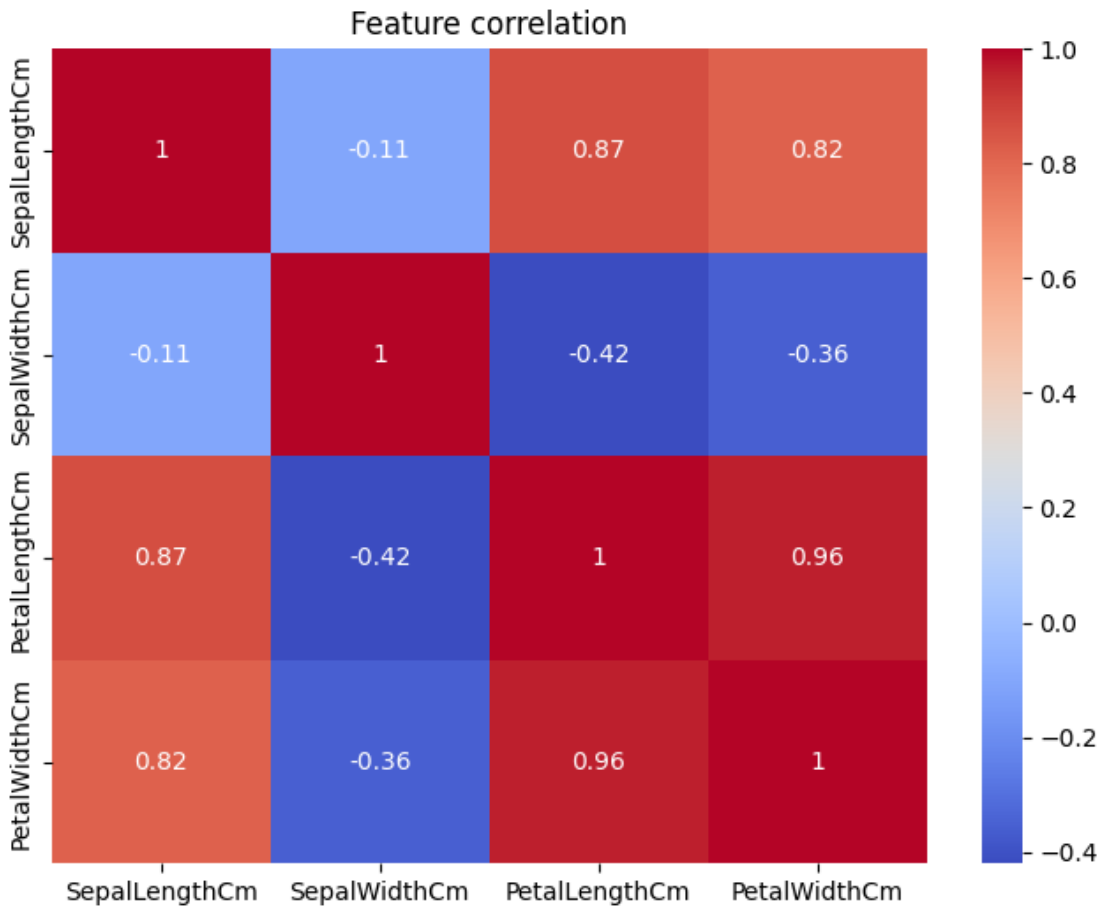
[42]:
```
plt.figure(figsize=(12, 8))

for i, col in enumerate(features, 1):
    plt.subplot(2, 2, i)
    sns.boxplot(x='Species', y=col, data=df)
    plt.title(col)

plt.tight_layout()
plt.show()
```



[43]:
```
sns.pairplot(df.drop('Id', axis = 1), hue='Species')
plt.show()
```



[44]:
```
plt.figure(figsize=(8,6))
sns.heatmap(df.drop(['Species', 'Id'], axis=1).corr(),
            annot=True, cmap='coolwarm')
plt.title("Feature correlation")
plt.show()
```



3 Hand Written Character Recognition

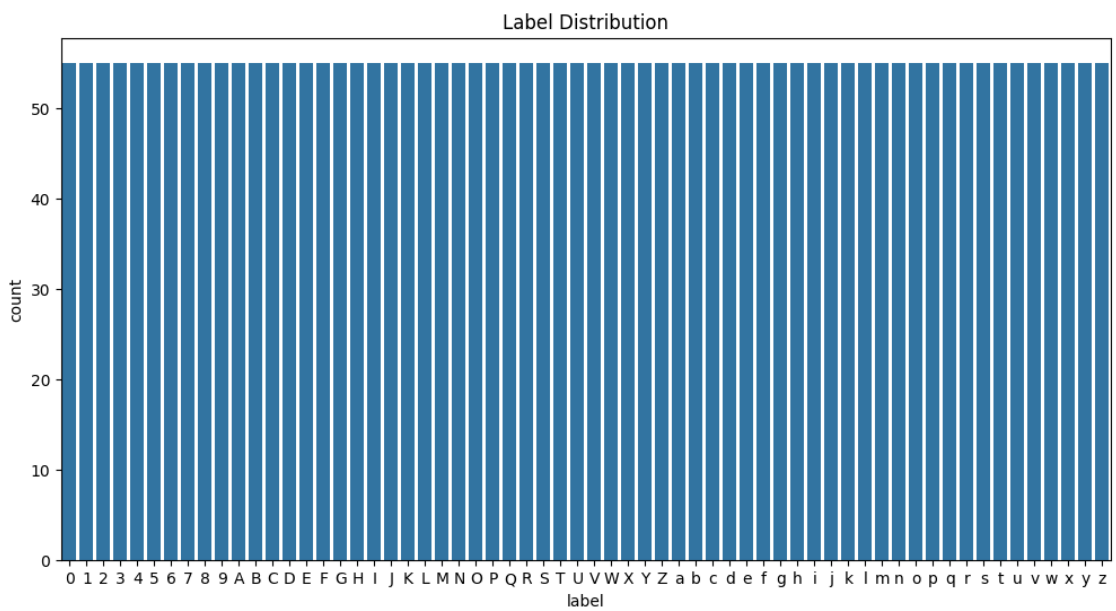
[45]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import os
import math

df = pd.read_csv("datasets/english.csv")
```

[46]:
```
print("Shape:", df.shape)
print("Info\n",df.info())
print("Head\n",df.head())

print(df.isnull().sum())
```

```
Shape: (3410, 2)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3410 entries, 0 to 3409
```

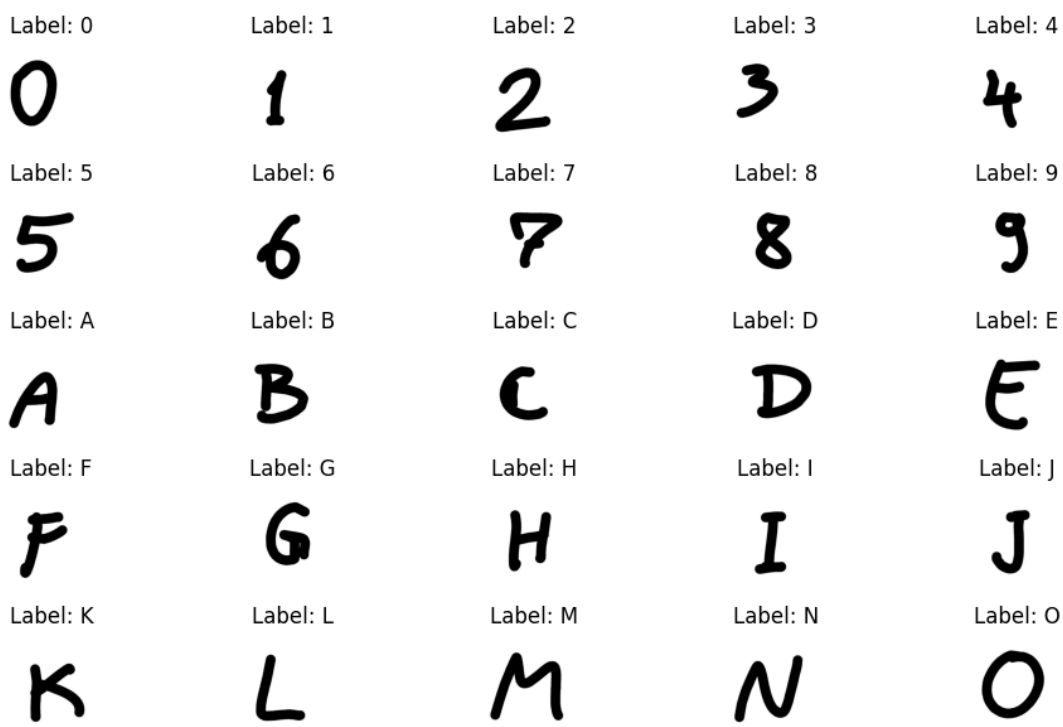
```
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   image   3410 non-null   object
 1   label   3410 non-null   object
dtypes: object(2)
memory usage: 53.4+ KB
Info
 None
Head
                 image label
0  Img/img001-001.png     0
1  Img/img001-002.png     0
2  Img/img001-003.png     0
3  Img/img001-004.png     0
4  Img/img001-005.png     0
image    0
label    0
dtype: int64
```

[47]:
```
plt.figure(figsize=(12,6))
sns.countplot(data=df, x="label", order=df["label"].value_counts().index)
plt.title("Label Distribution")
plt.show()
```



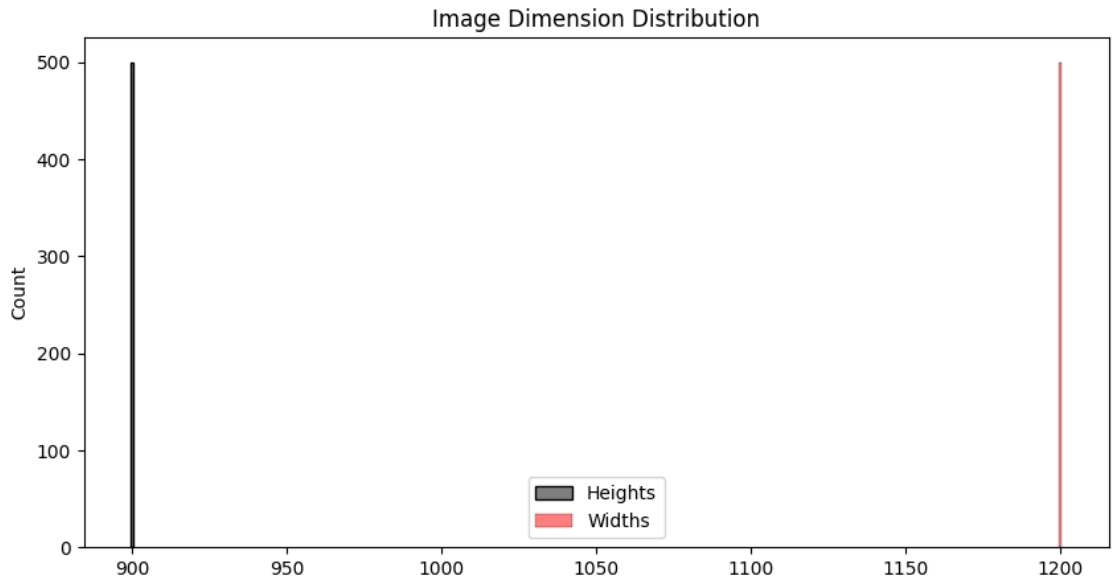
[48]:
```
from PIL import Image
```

```
plt.figure(figsize=(10, 6))

unique = df.groupby('label').first().reset_index().head(25)

for i, row in unique.iterrows():
    img_path = row['image']
    label = row['label']

    img = Image.open(f"datasets/{img_path}")

    plt.subplot(5, 5, i+1)
    plt.imshow(img, cmap='gray')
    plt.title(f"Label: {label}")
    plt.axis('off')

plt.tight_layout()
plt.show()
```



[49]:
```
from PIL import Image

heights = []
widths = []
```

```
for img_path in df['image'][:500]:
    img = Image.open(f"datasets/{img_path}")
    w, h = img.size
    widths.append(w)
    heights.append(h)


plt.figure(figsize=(10,5))
sns.histplot(heights, kde=True, color="black", label="Heights")
sns.histplot(widths, kde=True, color="red", label="Widths")
plt.legend()
plt.title("Image Dimension Distribution")
plt.show()
```



[50]:
```
import pandas as pd

df = pd.read_csv("datasets/diabetes.csv")
```

[51]:
```
print("Shape:", df.shape)
print("Info\n", df.info())
print("Head\n", df.head())
```

```
Shape: (768, 9)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
```

```
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
Info
 None
Head
    Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
```

[52]:
```
print(df.isnull().sum())
```

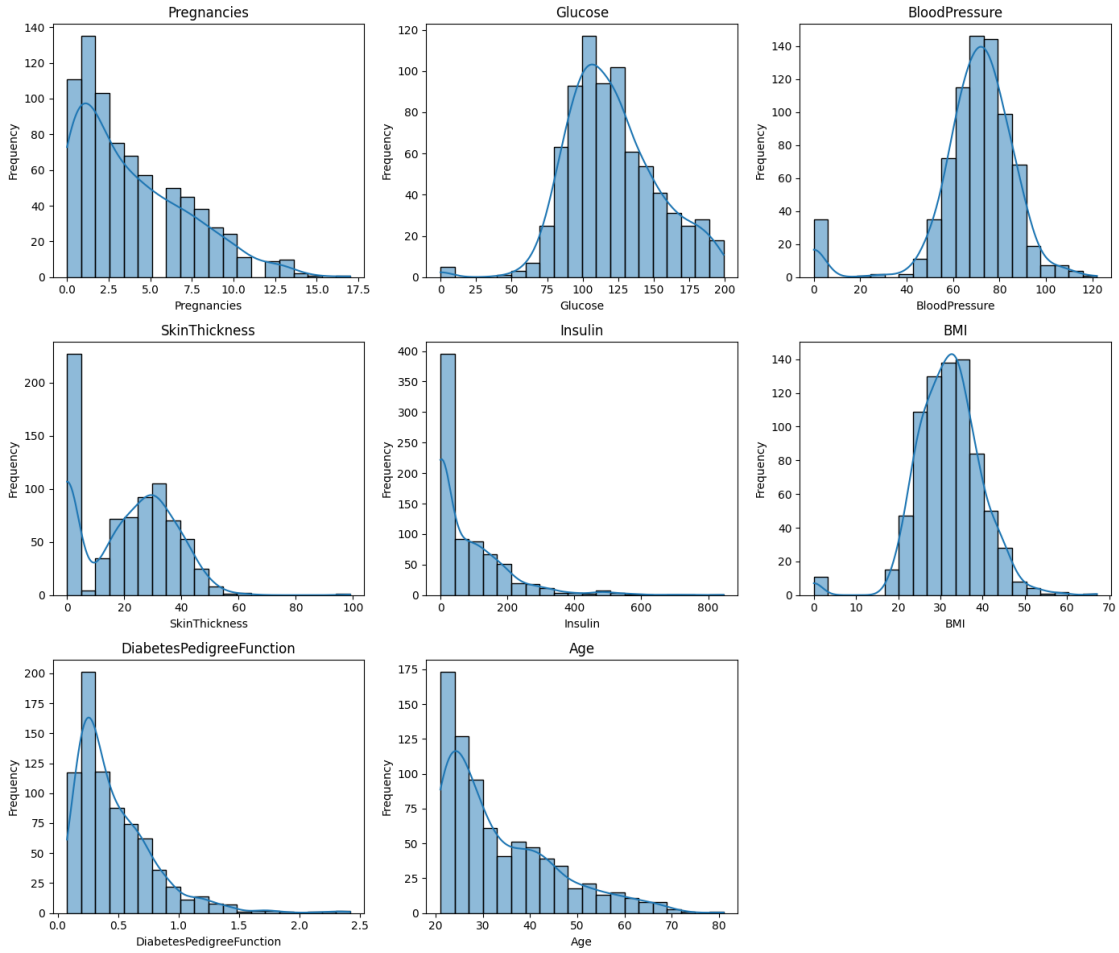
```
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64
```

[53]:
```
print(df.describe())
```

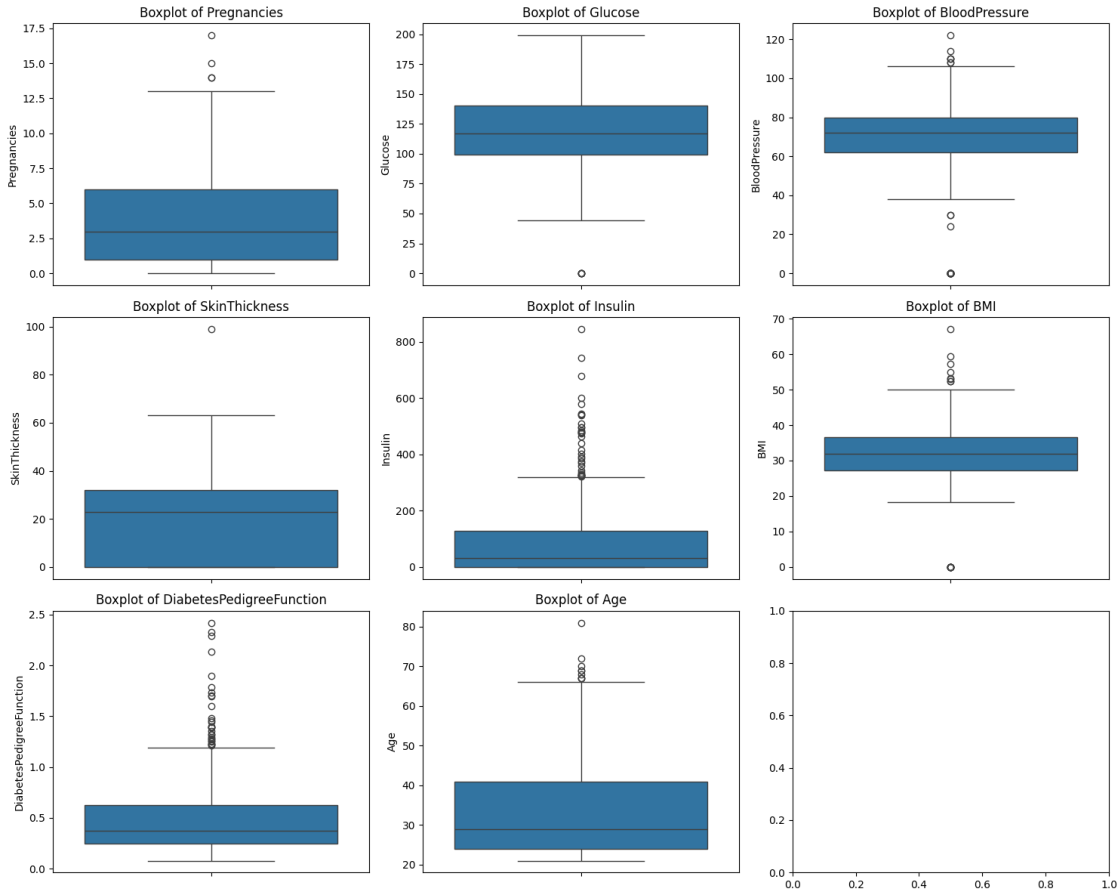
```
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   768.000000  768.000000     768.000000     768.000000  768.000000
mean      3.845052  120.894531      69.105469      20.536458   79.799479
std       3.369578   31.972618      19.355807      15.952218  115.244002
min       0.000000    0.000000       0.000000       0.000000    0.000000
25%       1.000000   99.000000      62.000000       0.000000    0.000000
```

```
50%       3.000000  117.000000      72.000000      23.000000   30.500000
75%       6.000000  140.250000      80.000000      32.000000  127.250000
max      17.000000  199.000000     122.000000      99.000000  846.000000

              BMI  DiabetesPedigreeFunction         Age     Outcome
count  768.000000                768.000000  768.000000  768.000000
mean    31.992578                  0.471876   33.240885    0.348958
std      7.884160                  0.331329   11.760232    0.476951
min      0.000000                  0.078000   21.000000    0.000000
25%     27.300000                  0.243750   24.000000    0.000000
50%     32.000000                  0.372500   29.000000    0.000000
75%     36.600000                  0.626250   41.000000    1.000000
max     67.100000                  2.420000   81.000000    1.000000
```

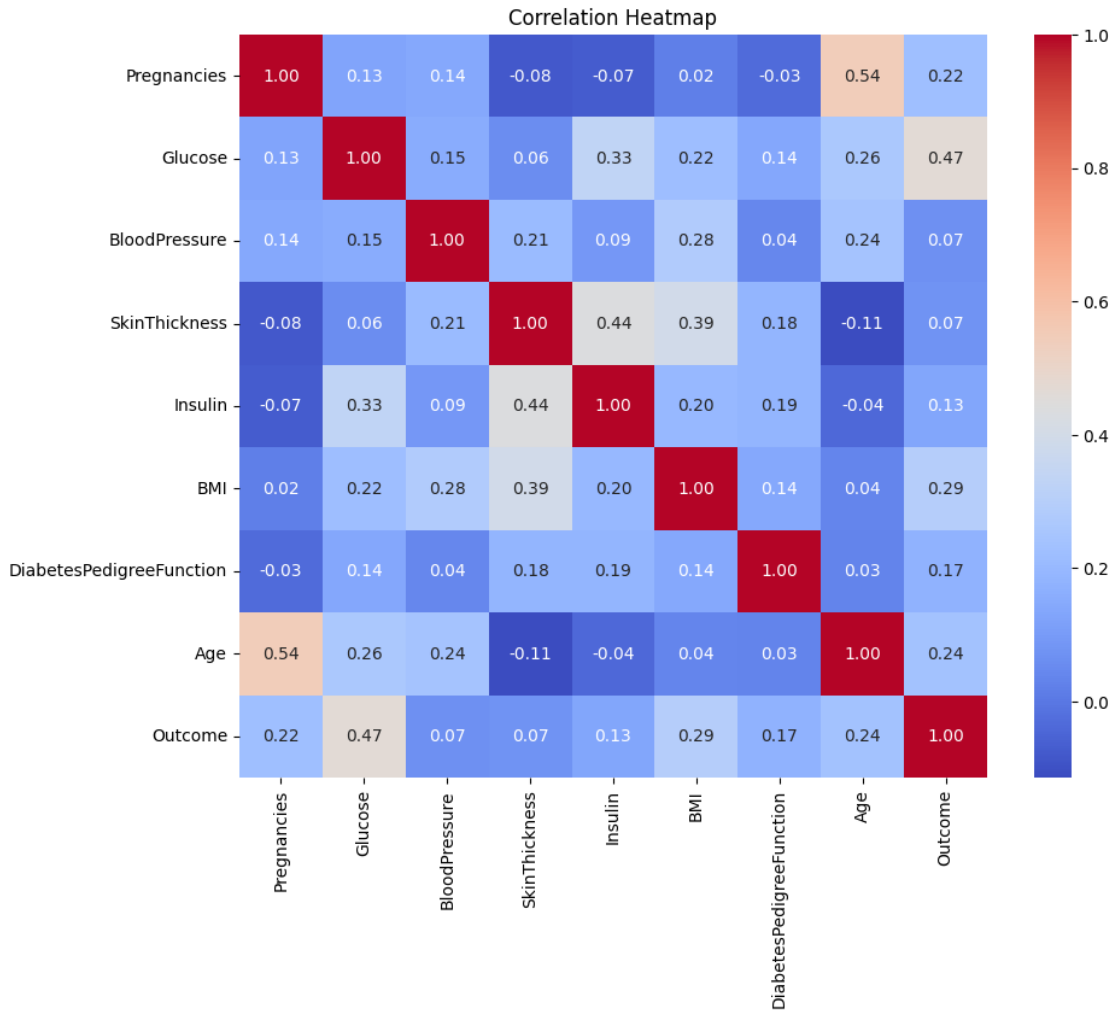
[54]:
```
features = df.drop('Outcome', axis=1).columns

plt.figure(figsize=(14, 12))

for i, col in enumerate(features, 1):
    plt.subplot(3, 3, i)
    sns.histplot(df[col], bins=20, kde=True)
    plt.title(col)
    plt.xlabel(col)
    plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```



[55]:
```
fig, axes = plt.subplots(3, 3, figsize=(15, 12))
axes = axes.flatten()

for i, col in enumerate(features):
    sns.boxplot(y=df[col], ax=axes[i])
    axes[i].set_title(f'Boxplot of {col}')


plt.tight_layout()
plt.show()
```



[56]:
```
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
```



4 Email Classification

[57]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv("datasets/email.csv")
df.head()
```

[57]:

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

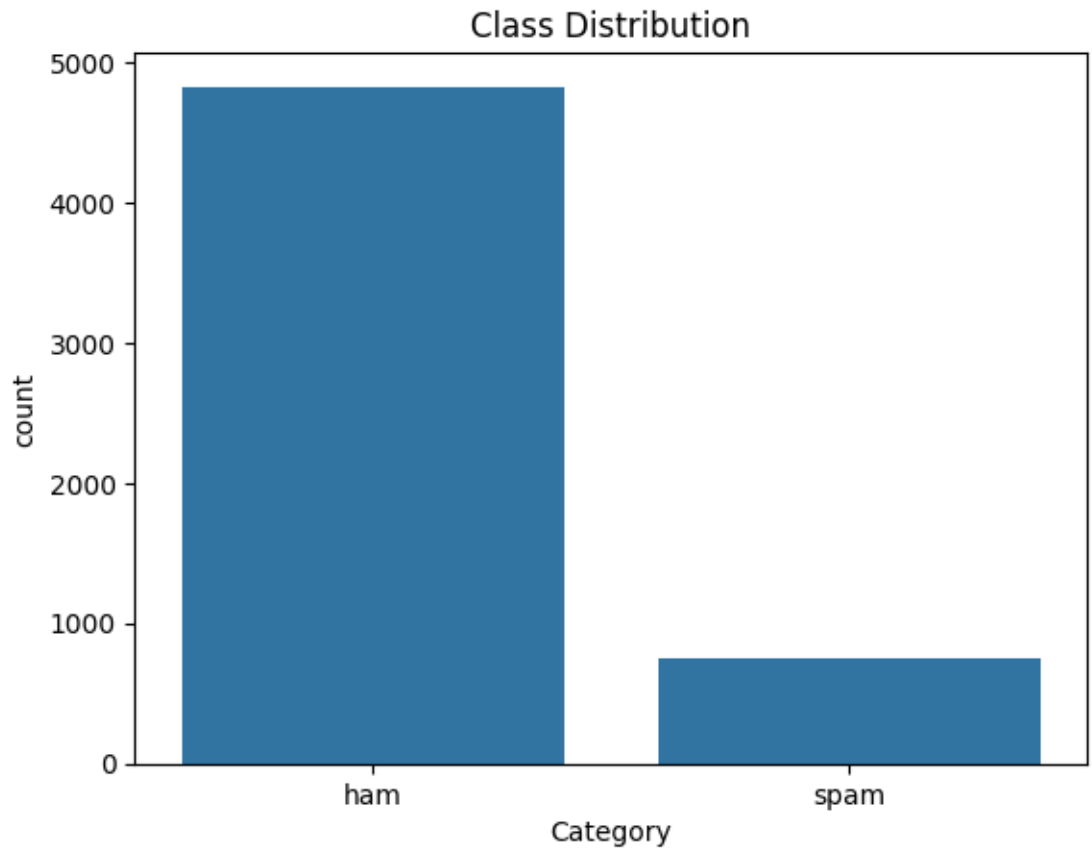
[58]:
```
print("Info\n", df.info())
print("Describe \n", df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5572 entries, 0 to 5571
Data columns (total 2 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Category  5572 non-null   object
 1   Message   5572 non-null   object
dtypes: object(2)
memory usage: 87.2+ KB
Info
 None
Describe
        Category                 Message
count      5572                    5572
unique        2                    5157
top         ham  Sorry, I'll call later
freq       4825                      30
```

[59]:
```
print("Null Values\n", df.isnull().sum())
print("Counts of spam and Ham \n", df['Category'].value_counts())
```

```
Null Values
 Category    0
Message     0
dtype: int64
Counts of spam and Ham
 Category
ham     4825
spam     747
Name: count, dtype: int64
```

[60]:
```
sns.countplot(x='Category', data=df)
plt.title("Class Distribution")
plt.show()
```



[61]:
```
import re

def clean_text(text):
    text = text.lower()
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"[^a-z\s]", "", text)
    return text

df['clean_text'] = df['Message'].apply(clean_text)
```

[62]:
```
df['word_count'] = df['clean_text'].apply(lambda x: len(x.split()))
df['char_count'] = df['clean_text'].apply(len)
```

[63]:
```
df.head()
```

[63]:
```
  Category                                            Message  \
0      ham  Go until jurong point, crazy.. Available only ...
1      ham                      Ok lar... Joking wif u oni...
2     spam  Free entry in 2 a wkly comp to win FA Cup fina...
3      ham  U dun say so early hor... U c already then say...
```

```
4      ham  Nah I don't think he goes to usf, he lives aro...

                                          clean_text  word_count  char_count
0  go until jurong point crazy available only in ...          20         102
1                             ok lar joking wif u oni           6          23
2  free entry in  a wkly comp to win fa cup final...          25         124
3        u dun say so early hor u c already then say          11          43
4  nah i dont think he goes to usf he lives aroun...          13          59
```

[64]:
```python
from collections import Counter

all_words = " ".join(df['clean_text']).split()
common_words = Counter(all_words).most_common(20)
common_words
```

[64]:
```
[('i', 2249),
 ('to', 2248),
 ('you', 2128),
 ('a', 1448),
 ('the', 1331),
 ('u', 1154),
 ('and', 971),
 ('is', 892),
 ('in', 889),
 ('me', 791),
 ('my', 759),
 ('for', 711),
 ('your', 677),
 ('it', 622),
 ('of', 622),
 ('call', 578),
 ('have', 573),
 ('on', 536),
 ('that', 514),
 ('are', 490)]
```

[65]:
```python
spam_words = " ".join(df[df['Category']=="spam"]['clean_text']).split()
ham_words = " ".join(df[df['Category']=="ham"]['clean_text']).split()

print("Spam words most common \n\n", Counter(spam_words).most_common(15))
print("\nNot Spam common words \n", Counter(ham_words).most_common(15))
```

```
Spam words most common

 [('to', 686), ('a', 380), ('call', 347), ('you', 287), ('your', 263), ('free',
219), ('for', 204), ('the', 201), ('now', 190), ('or', 188), ('is', 157), ('u',
153), ('txt', 151), ('on', 145), ('ur', 144)]
```

```
Not Spam common words
 [('i', 2195), ('you', 1841), ('to', 1562), ('the', 1130), ('a', 1068), ('u',
1001), ('and', 849), ('in', 817), ('me', 761), ('my', 748), ('is', 735), ('it',
594), ('of', 525), ('for', 507), ('that', 488)]
```

Results and Discussions:

Dataset	**Type of ML Task**	**Suitable ML Algorithm**
Iris Dataset	Classification	K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree
Loan Amount Prediction	Regression	Linear Regression, Decision Tree Regressor, Random Forest Regressor
Diabetes Prediction	Classification	Logistic Regression, Support Vector Machine, Random Forest
Email Spam Detection	Text Classification	Naive Bayes, Support Vector Machine, Logistic Regression
Handwritten Character Recognition	Image Classification	Convolutional Neural Network (CNN), Support Vector Machine

Learning Practices:

- Understand the fundamentals of data analysis using Python libraries such as NumPy, Pandas, Matplotlib, and Seaborn.
- Analyze datasets using descriptive statistics and exploratory data analysis techniques.
- Visualize data distributions and relationships using appropriate plots and charts.
- Identify suitable machine learning tasks such as classification and regression for different datasets.